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Course Based Project

Power Systems Engineering (BEEE203L)

POWER THEFT ANALYZER

To Professor Jacob Raglend

by -

Arvind Rajan S (24BEE0162)

Abhinav Krishnan A (24BEE0136)

SCOPE

This project focuses on detecting power theft in an electrical distribution network using a simple mathematical. The system monitors energy flow at three important points in the power system: the Feeder level, the Distribution Transformer level, and the Consumer level. By comparing the energy entering a section of the network with the energy recorded by all meters in that section, the system can calculate the loss percentage. If the loss exceeds a calculated threshold, it generates an alert indicating possible theft. The project aims to identify ten different methods of power theft that are commonly observed in Tamil Nadu, including pre-meter tapping on poles, meter bypass, feeder tap connections, illegal transformers, and neutral tampering tricks. A hardware prototype has been built using an ESP32 microcontroller, a small transformer, two loads (one legal and one representing theft), voltage and current sensors, and a toggle switch. This prototype simulates a pre-meter theft scenario and demonstrates real-time detection of illegal current draw. The data from the prototype is sent wirelessly over Wi-Fi to a website dashboard. The website displays live electrical parameters such as voltage, total current, legal load current, power, and a theft status indicator. It also includes pages for viewing alerts and analytics. The project proves that a low-cost monitoring system can effectively identify power theft at different stages of the distribution network without requiring heavy computational infrastructure.

LITERATURE SURVEY

Tamil Nadu Power Sector Structure:

In Tamil Nadu, the electricity sector is managed by several entities under the umbrella of the Tamil Nadu Electricity Board. TANGEDCO handles distribution, TANTRANSCO handles transmission, TNPDC and TNPGL are involved in power generation projects, and TNGEL looks after green energy initiatives.

Power System Flow from Generation to Consumer:

The electricity supply chain follows a standard power system structure. It begins at the Generation Station where power is produced. From there, it travels through Transmission Lines at very high voltage to reduce losses. The power then reaches a Grid Substation where voltage is stepped down. From the substation, Primary Feeders operating at 11 kV or 33 kV carry power to different areas. Industries often receive power at this voltage level using Current Transformers and Potential Transformers along with High Tension metering panels. The next step is the Distribution Transformer which steps down the voltage further. From the transformer, Secondary Feeders carry power on Low Tension lines at 400 Volts for three-phase supply and 230 Volts for single-phase supply. A Service Drop connects these lines to individual buildings. Finally, the power passes through a Meter and Service Head before entering the Consumer's Internal Wiring.

Types of Losses in the System:

The losses in this system are broadly divided into two categories.

Commercial Loss occurs due to theft, meter tampering, or billing inefficiencies. Technical Loss occurs naturally due to the resistance of wires.

Evolution of Metering Technology:

Over the years, the meters used at consumer premises have evolved. Older Single Phase Electromechanical Meters have a rotating disc and are vulnerable to mechanical tampering. Single Phase Electronic Meters provide better accuracy but can still be bypassed or manipulated. The latest technology is the Smart Meter. Smart meters offer several benefits. No one needs to visit homes to take readings. Power theft can be detected almost instantly. Consumers can view their usage online or through a mobile app. Utilities can manage the power grid more effectively using real-time data.

Ten Common Methods of Power Theft:

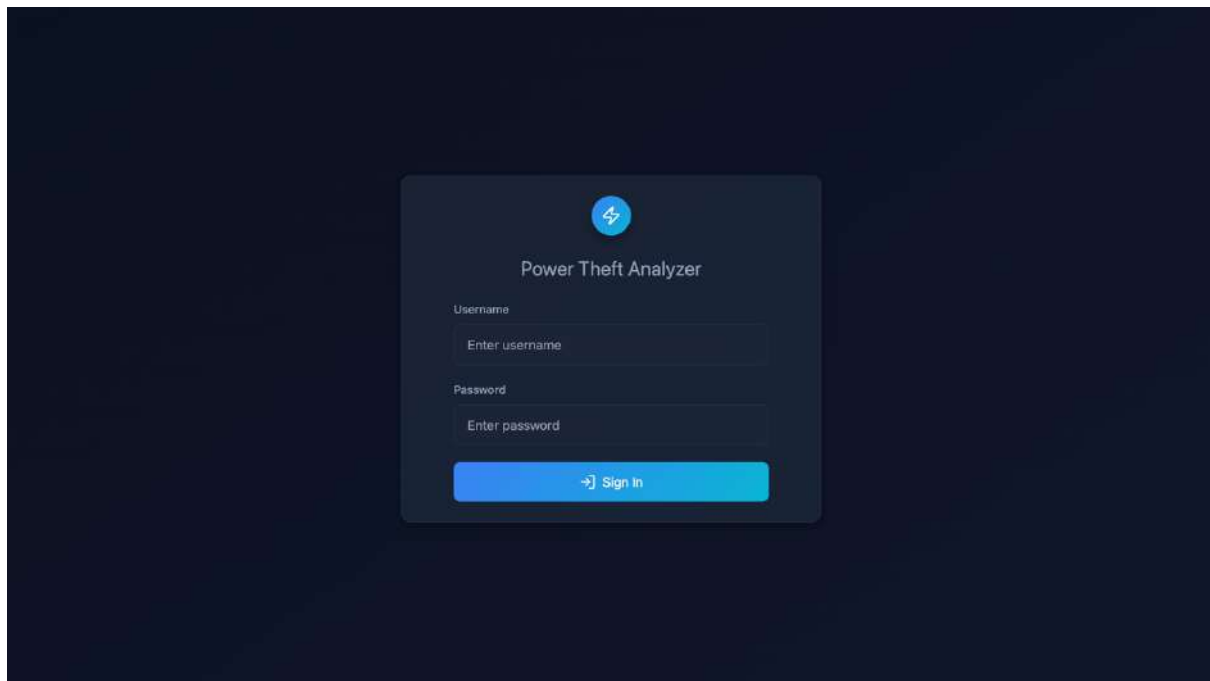
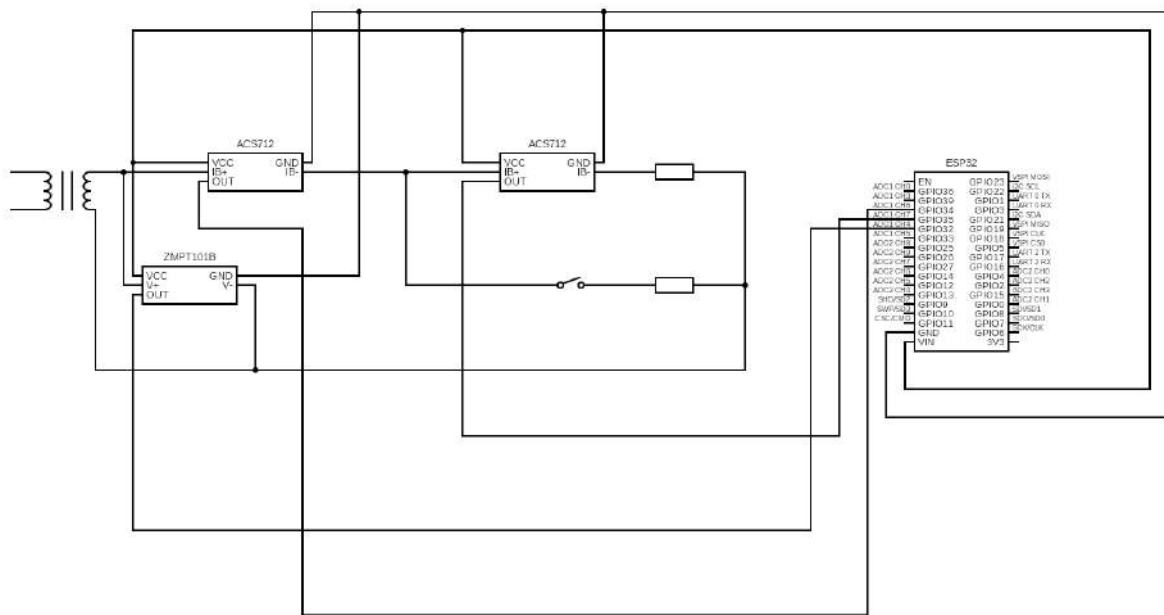
The project also studies the ten most common methods used for power theft. The first method is tapping on the pole before the meter, which draws power directly from the service line without passing through the meter. The second method is a bypass inside the meter box using jumper wires. The third method involves physically tampering with an old meter to slow down the recording disc or reset the reading. The fourth method is an illegal connection made directly at the feeder tap point on the high voltage line. The fifth method occurs in multi-tenant buildings where tapping is done inside the distribution room before individual meters. The sixth method targets High Tension metering panels by bypassing Current Transformers or Potential Transformers to under-report industrial consumption. The seventh method involves setting up an illegal

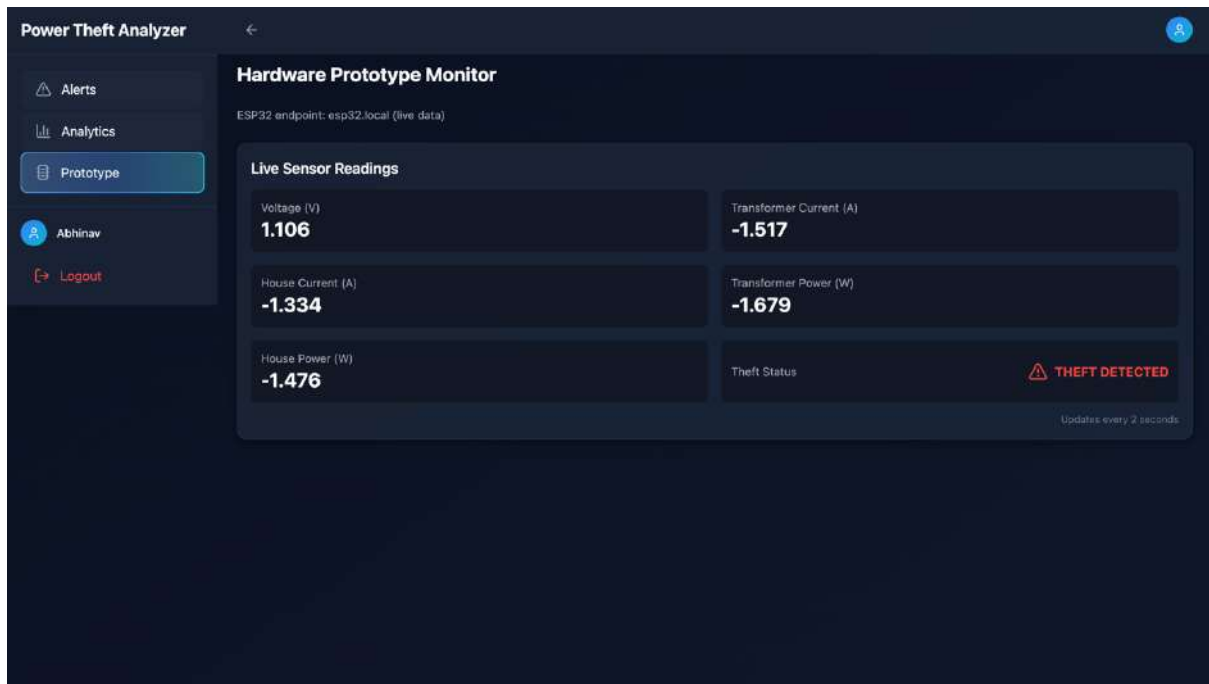
transformer in a remote area to draw power directly from the feeder. The eighth method uses a neutral disconnect switch to manipulate meter readings. The ninth method is tapping a neighbor's phase wire so that the neighbor's meter records the extra usage instead of the thief's meter. The tenth method involves disconnecting the neutral wire or using earthing tricks to create an imbalance that the meter cannot accurately record.

Government of Tamil Nadu Anti-Theft Strategy:

The Government of Tamil Nadu, through TANGEDCO, has adopted a two-part strategy to reduce power theft. The first part involves large-scale deployment of technology, specifically smart meters, under the Government of India's Revamped Distribution Sector Scheme known as RDSS. The second part is active enforcement through dedicated anti-power theft squads that conduct surprise raids and inspections. The official strategy works by monitoring the total energy flowing through a feeder or transformer and comparing it with the sum of all consumer meter readings downstream. This comparison helps pinpoint areas with high Aggregate Technical and Commercial losses, which indicate potential large-scale theft or the presence of illegal connections. The project described in this report follows a similar principle but applies it in a structured, mathematical way at three distinct levels using a prototype that simulates real-world conditions.

Block Diagram, Software & Hardware overview:





Power Theft Analyzer

Alerts

Analytics

Prototype

Abhinav

Logout

Alerts Management

Status: All

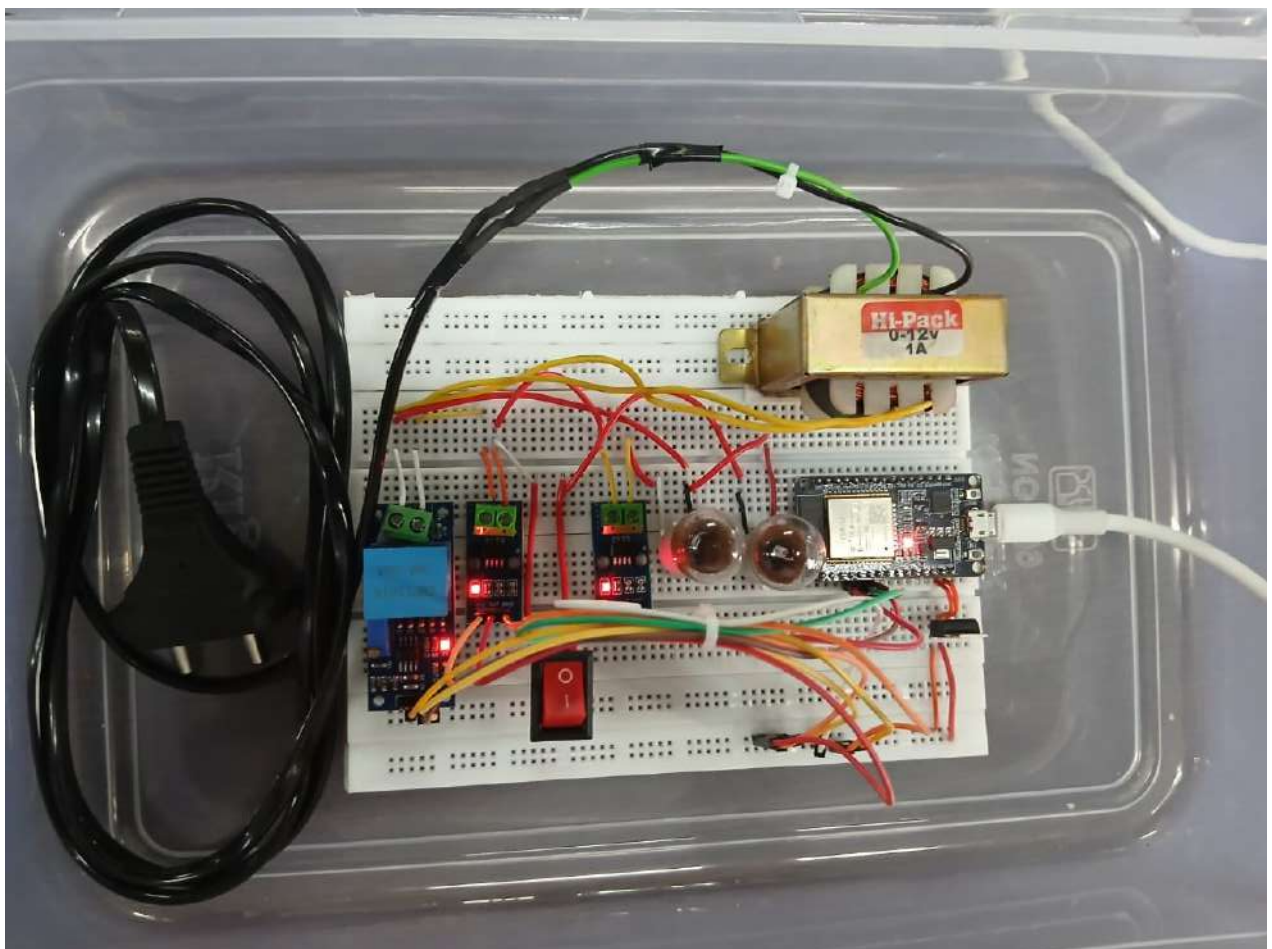
Total: 10 alerts

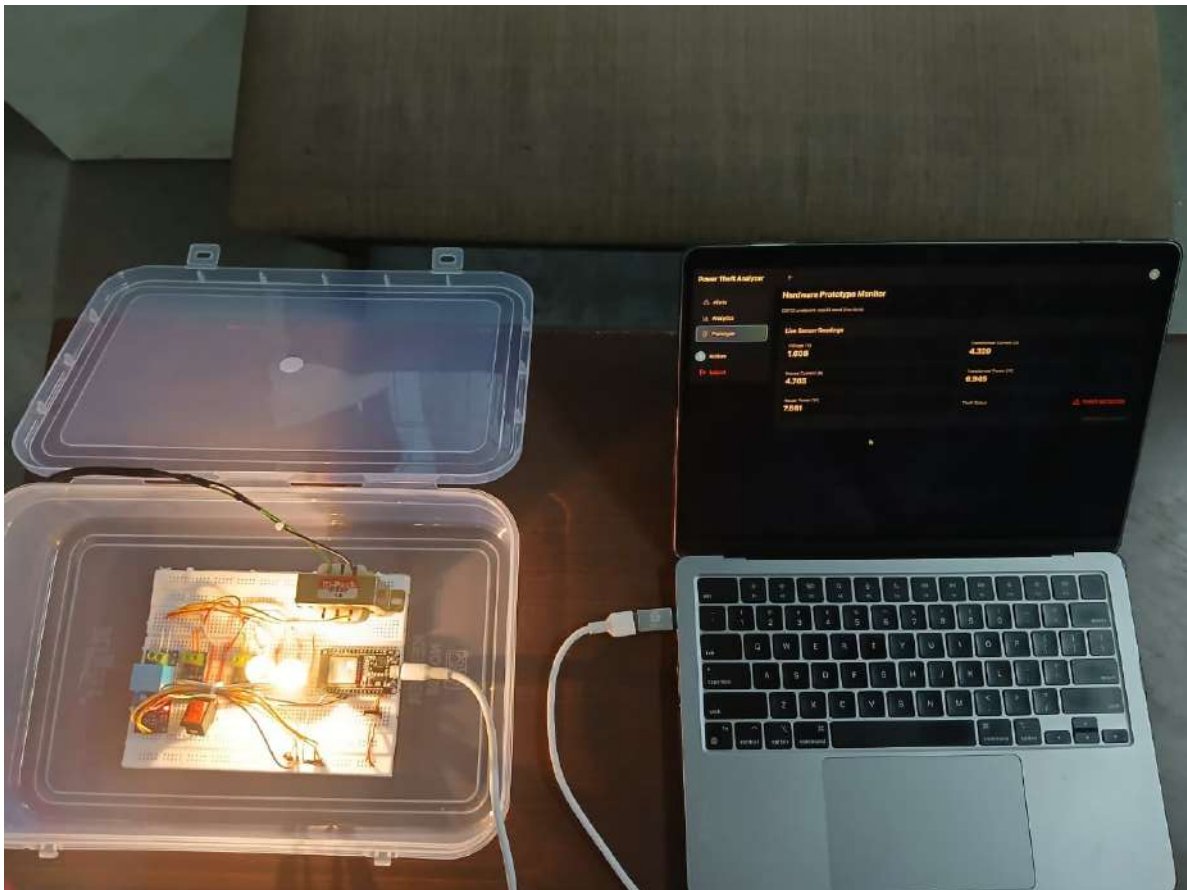
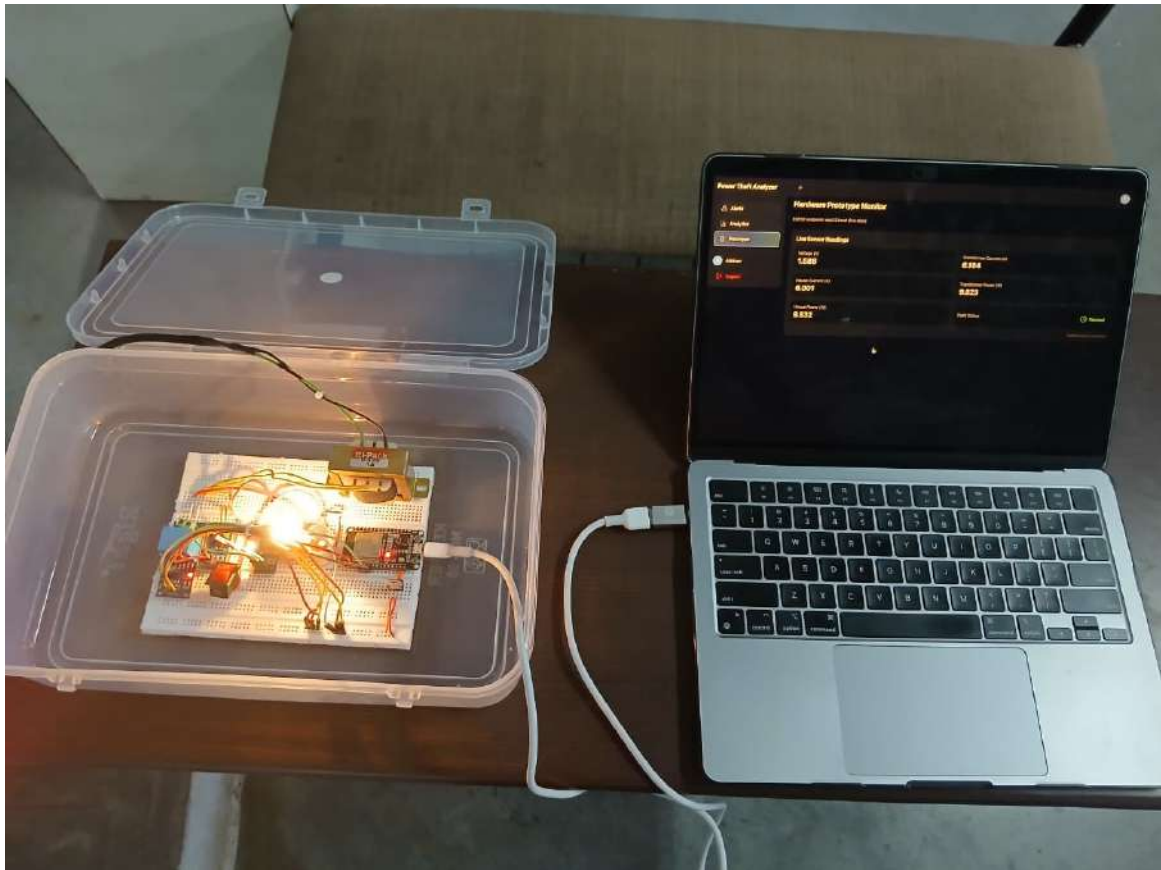
Meter ID	Location	Status	Actions
meter-2530	Zone 254	investigating	Investigating
meter-2032	Zone 204	new	New
meter-2084	Zone 209	investigating	Investigating
meter-2783	Zone 271	investigating	Investigating
meter-2032	Zone 204	investigating	Investigating
meter-2483	Zone 241	new	New
meter-2212	Zone 222	investigating	Investigating
meter-2579	Zone 268	investigating	Investigating
meter-2792	Zone 280	new	New
meter-2999	Zone 300	new	New

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METHODOLOGY

System-Level Detection Methodology:

The system monitors energy flow at three levels: Feeder meter at substation, Transformer meter on primary side, and Consumer meters at premises. Readings are sent to a central server every 15 to 60 minutes. Detection uses five mathematical rules.

Rule A (Transformer Balance): Compares transformer input with sum of consumer meters. Loss above 10-15% triggers alert. Detects pole tapping and building room theft.

Rule B (Feeder Balance): Compares feeder input with sum of transformer meters. Loss above 6-10% triggers alert. Detects HT tapping and illegal transformers.

Rule C (Tamper Events): Meter firmware detects cover open, magnet, phase reversal, neutral missing, and current imbalance. Immediate alert.

Rule D (Neighbor Comparison): Usage below 50% of nearby homes indicates bypass.

Rule E (Historical Comparison): Usage below 30% of same-hour average over past 4 weeks indicates recent tampering.

Prototype Implementation Methodology:

The prototype simulates single-phase pre-meter theft. A step-down transformer (230V AC to 12V AC) safely represents the distribution transformer source. Two loads connect to the circuit: a legal load (normal consumer) and a thief load controlled by a toggle switch. Closing the switch draws current bypassing the legal load current sensor, exactly mimicking Method 1 theft (pre-meter tapping).

Three sensors measure electrical parameters: one voltage sensor monitors supply voltage continuously. Two current sensors: first measures total current from transformer (legal + thief), second measures only legal load current on its branch.

An ESP32 microcontroller reads analog sensor outputs, converts to digital, and calculates voltage, current, and active power. It takes readings every two seconds and compares total current to legal current. If total current exceeds legal current by a set margin, firmware flags possible theft.

The ESP32 connects to a Wi-Fi network (mobile hotspot in testing) and runs an embedded web server at a fixed local address. The server returns voltage, total current, legal current, power, and theft status flag in a simple data format.

A companion website has three pages: Prototype, Alerts, Analytics. The Prototype page sends requests to the ESP32 address every two seconds, fetches data, and updates displayed values. The website communicates with a backend server storing alerts in a SQLite database (tables: meters, readings, alerts). When ESP32 indicates theft, the website notifies the backend to create an alert entry.

The prototype does not implement Part 1 transformer/feeder balance rules (single load point only). It validates the principle of comparing current flow before and after a tap point.

COMPONENTS USED AND COST

- ESP32 Development Board – 300 Rs.
- Rocker Switch - 14 Rs.
- Wires - 30 Rs.
- Screw Driver - 45 Rs.
- Wire Cutter - 60 Rs.
- Voltage Sensor Module – 199 Rs.
- Current Sensor Module (ACS712) – (2 units) – 200 Rs.
- 10k ohms Resistors - 10 Rs.
- Step-Down Transformer (230V to 12V) – 150 Rs.
- Insulation Tape - 15 Rs.
- Loads (Bulbs/Resistors) – 50 Rs.
- Breadboard - 60 Rs.
- USB Type B cable - 50 Rs.
- Type A to Type C converter - 30 Rs.

Total Estimated Prototype Cost: 2563 Rs.

OUTCOMES

The prototype successfully detects theft in real time. When the theft switch is off, both current sensors read similar values. When the switch is on, total current becomes larger than legal current. The ESP32 detects this and updates theft status. The website displays live voltage, total current, legal current, power, and a red/green theft indicator. Data flows automatically from hardware to website every two seconds. Alerts are stored in the SQLite database. The system proves that low-cost components can reliably detect pre-meter tapping.

NOVELTY

- Uses simple arithmetic rules, not complex AI or ML.
- Hardware built with low-cost, easily available components.
- Maps ten specific theft methods to five detection rules.
- Provides real-time web dashboard for monitoring.
- Aligns with TANGEDCO and RDSS strategy.

FUTURE SCOPE

- Fix Alerts page to display real alerts from database.
- Improve Analytics page with detailed charts and trends.
- Scale up with cloud backend for multiple nodes.
- Refine thresholds using actual utility data.
- Expand prototype to three-phase simulation.
- Add mobile app for instant push notifications.
- Integrate with utility billing systems to prioritize enforcement raids.